

Experiment 01 a) Built-in API

Learning Objective: Integration of Built-in API into a Web page.

Tools:

Theory:

Steps to Integrate Gemini API in a Python Web Application

Step 1: Create a Google AI Studio Account

- Visit **Google AI Studio**.
- Sign in using your Google account.
- Generate a Gemini API Key.

Step 2: Install Required Packages

```
pip install flask google-generativeai python-dotenv
```

Step 3: Create a **.env** File

Store your API key securely.

```
GEMINI_API_KEY=your_api_key_here
```

Step 4: Create a Python Flask Application

- Create a Flask application to serve as the backend web server.
- Configure Flask routes for the web interface.

Step 5: Connect the Gemini API

- Import the Google Generative AI Python SDK.
- Configure the SDK using the Gemini API key.
- Create a Gemini model instance (for example, **gemini-2.5-flash**).

Step 6: Create a **/chat** Route

- Receive user input from the web page.
- Send the user's prompt to the Gemini API.
- Receive the AI-generated response.
- Return the response to the web page in JSON format.

Step 7: Create a Simple HTML Page

- Add a text box for user input.
- Add a **Send** button.
- Use JavaScript Fetch API to send the message to the Flask backend.
- Display the Gemini AI response on the webpage.

Step 8: Run the Application

`python app.py`

Step 9: Open the Application

Open your browser and visit: `http://127.0.0.1:5000`

Step 10: Test the Chatbot

- Enter a question in the text box.
- Click the **Send** button.
- The Gemini AI response will be displayed on the webpage.

Implementation:

OUTPUT:

Learning Outcomes: The student should have the ability to

Course Outcomes: Students will be able to learn:-

Conclusion:

Viva Questions:

1. What are built-in browser APIs, and how do they enhance the functionality of web applications?
2. Explain how the Geolocation API works. How do you request and display the user's current location on a webpage?
3. Compare `localStorage`, `sessionStorage`, and `cookies` in terms of scope, storage limit, and use cases.

4. What are the steps to show browser notifications using the Notification API, and how do you handle permission requests?
5. Discuss potential security and privacy concerns when using built-in APIs like Geolocation, Camera, or Microphone.

For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	Total
Marks Obtained				